Leveraging MOOCs for Individuals with Visual Impairments

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Abstract. Communication with individuals with special needs, particularly those affected by blindness, remains a challenge in contemporary society. This challenge underscores a growing societal concern to overcome barriers blocking equal opportunities. Leveraging new technologies and effective educational practices, computer applications offer a pathway for full digital access, empowering individuals with visual impairment according to their needs. In Portugal, Information and Communication Technology (ICT) advancements reshape educational roles, with MOOCs (Massive Open Online Courses) emerging as an alternative teaching method, particularly beneficial for ICT growth. However, while these platforms offer open access to learning, they often fail to consider the specific needs of learners with disabilities. The objective of this project is to develop an inclusive MOOC model that specifically addresses the needs of communities with visual impairments. The significance of this lies in promoting equal access to education and overcoming the exclusion faced by individuals with disabilities. This model will incorporate assistive technologies such as text-to-voice conversion for blind learners and other sensory-based adaptations, ensuring full engagement with the course content.

Keywords: Inclusive education · Visual impairment · Accessibility · MOOCs · Digital learning

1 Introduction

The digital transformation of education has made learning opportunities more accessible than ever. However, many online learning environments remain inaccessible to individuals with visual impairments [1–4]. Visual impairments pose barriers that limit opportunities for education, employment, and social integration. Technological advancements, though, offer a pathway toward inclusive learning environments.

The integration of Information and Communication Technologies (ICT) into education is reshaping learning dynamics. While MOOCs (Massive Open Online Courses) have emerged as a promising avenue for democratizing education, their lack of inclusivity for learners with disabilities presents a critical gap [1–4]. The current project seeks to address this gap by designing an inclusive MOOC model tailored to the needs of visually impaired learners.

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The proposed solution incorporates adaptive technologies and inclusive instructional design strategies to ensure accessibility and engagement for blind users. By fostering autonomy and digital literacy, this initiative aims to enhance the educational and professional opportunities available to visually impaired learners. This paper outlines the development of the pedagogical model, key methods employed and project outcomes.

2 Methodology

The development of an inclusive MOOC is rooted in universal design, adaptability, and user-centered learning principles, ensuring full accessibility for visually impaired learners. To develop an inclusive and accessible MOOC model for visually impaired learners, a structured and iterative approach was adopted. The methodology encompassed the following key steps:

1. Needs Assessment: This phase involved an extensive review of existing literature and consultations with visually impaired users and specialists. The objective was to identify specific challenges faced by blind learners in digital environments, including navigation difficulties, inaccessible course materials, and the absence of sensory-based alternatives.
2. Design and Development: During this stage, adaptive technologies and accessibility features were integrated into the MOOC platform. Written course materials were converted into high-quality audio formats using advanced text-to-speech technologies, ensuring comprehensive auditory engagement. Course materials were segmented into structured audio chapters, enabling learners to navigate through content efficiently. Interactive quizzes and exercises provided real-time verbal feedback to reinforce learning outcomes, while playback control options allowed users to adjust speed, pause, or skip content. Additionally, multilingual audio support was implemented to accommodate learners from diverse linguistic backgrounds.
3. Collaborative Design Approach: Collaboration played a crucial role in ensuring the inclusivity and practicality of the MOOC platform. A multidisciplinary team was assembled, including accessibility experts who provided insights into best practices, educators and content creators who adapted instructional materials for an audio-based format, technologists who developed and tested the assistive features, and organizations supporting the blind who validated the approach and facilitated user testing.
4. Iterative Testing and Feedback: Continuous user testing was conducted with visually impaired learners. Regular user feedback allowed for improvements in platform functionality, accessibility features, and overall user experience. Usability testing focused on ease of navigation, effectiveness of adaptive features, and learning outcomes, ensuring alignment with best practices in accessibility and education.
5. Evaluation: User experience testing measured the platform's usability and engagement levels, while learning outcomes were tracked through course completion rates and participant feedback. Continuous improvements were guided by data-driven insights, incorporating emerging technologies such as AI-driven voice assistants and haptic feedback to further enhance accessibility. Routine audits ensured compliance with global accessibility standards, including the Web Content Accessibility Guidelines (WCAG).

2.1 Concept

This project is founded on the principle that education should be accessible to all, regardless of physical limitations. While MOOCs have democratized learning by providing open access to educational content, they often fail to accommodate learners with disabilities, particularly those with visual impairments [1–4]. The proposed model seeks to bridge this accessibility gap by integrating assistive technologies that facilitate seamless interaction with digital content.

By leveraging advancements in ICT, the model incorporates adaptive features such as text-to-voice conversion. These elements are designed to foster autonomy, enabling visually impaired learners to engage with course materials independently. The inclusive pedagogical framework ensures that digital learning environments are not only accessible but also effective in delivering meaningful educational experiences [5–8].

This initiative goes beyond accessibility; it aims to promote digital literacy, social inclusion, and professional empowerment for individuals with visual impairments. By addressing current limitations in MOOCs, the project aspires to create a more equitable and inclusive digital education landscape.

2.2 Objectives

The primary objective of this project is to develop an inclusive MOOC model that specifically addresses the educational needs of visually impaired learners, ensuring equal access to digital learning environments. This initiative aims to eliminate the barriers that often hinder individuals with visual impairments from fully engaging with online education platforms.

To achieve this, the project focuses on the following key objectives:

- Enhance Digital Accessibility—Develop a MOOC framework that integrates assistive technologies, such as text-to-voice conversion, screen readers, and sensory-based adaptations to facilitate seamless navigation and content interaction for blind learners.
- Promote Equal Educational Opportunities—Ensure that individuals with visual impairments have the same access to MOOC content as their sighted peers, reducing exclusion and fostering academic participation.
- Leverage ICT for Inclusive Learning—Utilize ICT to create a fully accessible digital learning environment that supports autonomy and digital literacy among visually impaired learners.
- Develop an Inclusive Pedagogical Model—Design an instructional framework that integrates universal design principles, making MOOCs not only accessible but also engaging and effective for learners with disabilities.
- Bridge the Digital Divide—Address the current accessibility gaps in MOOCs, ensuring that online education platforms cater to diverse learners, particularly those with visual impairments.
- Encourage Professional and Social Inclusion—Empower visually impaired individuals by expanding their educational and professional opportunities, ultimately supporting their integration into the workforce and society.

By integrating assistive technologies and inclusive instructional strategies, this project aims to set a new standard for accessibility in digital education, making MOOCs more inclusive and equitable for all learners.

2.3 System Architecture

To enhance inclusivity, MOOCs integrate assistive technologies, employing text-to-speech (TTS) systems to facilitate seamless translation between written content and auditory output. This technology ensures that blind learners can access digital course content, while also maintaining a visually engaging environment for sighted participants. The courses further incorporate multimedia elements, such as audio descriptions, verbalized image content, and tactile adaptations, creating a fully immersive and inclusive learning experience [5–8].

The MOOCs employ a research-driven pedagogical model that integrates an automatic translation system powered by text-to-voice conversion. This system supports multimodal interaction, converting text to speech and ensuring that blind learners can navigate course materials independently. This model consists of:

- A digital content repository (DB) that stores educational materials in accessible formats.
- Content experts (teachers) who curate and present materials using tools like Word, PowerPoint, and multimedia content, ensuring audio-supported accessibility.
- Screen readers and speech synthesis tools, which enable the system to interpret text-based content and convert it into structured audio output.
- Audio-enhanced interfaces, responsible for guiding blind learners through course navigation, ensuring equal access to all learning materials.

This architecture eliminates the need for specialized programming knowledge or familiarity with complex accessibility tools, making the platform highly user-friendly for both teachers and students.

A crucial component of these MOOCs is the text-to-speech adaptation system, specifically designed to address the distinct learning needs of both blind and sighted individuals in an educational setting.

The architecture consists of two key modules:

1. Text-to-Speech Module—Converts written text into synthesized voice output, allowing blind learners to access content through audio narration.
2. Speech-to-Text Module—Enables voice commands and dictation, allowing blind learners to interact with course materials through spoken input.

By leveraging advanced speech recognition and audio-based navigation technologies, this system ensures that blind learners can engage naturally with digital content, fostering a fully immersive and interactive learning experience.

3 Evaluation

The assessment procedure for the proposed architecture and MOOC content was implemented to test its efficiency, effectiveness, usability, and accessibility features for blind learners. The Quality of Experience Framework (QEF) was applied to measure the quality of the educational content, ensuring that the platform meets the needs of blind users. This process involved collecting user feedback through questionnaires and analyzing the suitability of the MOOC for audio-based learning [6–9].

3.1 The QEF Approach for Blind Learners

QEF has been used for three decades to assess the quality of educational software products during their development cycle. It ensures consistency throughout the evaluation process and helps determine whether the MOOC effectively meets accessibility requirements for blind students. The framework uses performance indicators based on the ISO 9126 and SCORM standards to provide a quantitative representation of product quality [6–9]. The assessment focuses on three key dimensions:

- Pedagogical: Evaluates content structure, accessibility, and user engagement.
- Ergonomic: Assesses usability, accessibility, and adaptability of audio-based learning tools.
- Management: Ensures that educational content for blind learners is developed and validated by experts.

During the usability testing, 21 participants, including six blind users, interacted with a lesson in the inclusive MOOC prototype. These users then completed a five-point Likert scale questionnaire to evaluate the usability, adaptability, and content quality.

3.2 Preliminary Findings & Future Steps

The results showed that blind users expressed high satisfaction with the MOOC's audio features, with over 57% selecting "strongly agree" on key usability and accessibility metrics. Participants highlighted the effectiveness of audio descriptions, screen reader compatibility, and the flexibility to start and stop lessons at will.

However, areas for future improvement include enhancing the pedagogical evaluation component, which scored lower than other assessed dimensions. Future iterations of the MOOC should focus on optimizing instructional design to ensure that audio-based content is engaging, structured, and interactive. This includes:

- Incorporating more detailed audio descriptions of visual content, ensuring that blind users receive comprehensive information about diagrams, images, and animations through descriptive narration.
- Implementing an interactive learning approach, where users can engage with content through voice commands or touch gestures rather than passive listening.
- Developing adaptive learning paths, allowing blind learners to navigate courses at their own pace with personalized feedback based on their performance.

Also, while the existing MOOC platform incorporates audio features to support blind learners, these functionalities remain limited in scope. Current accessibility relies heavily on screen reader compatibility and static TTS conversion, which, while functional, does not fully enable independent and interactive learning. Blind learners often face challenges in navigating course content, engaging with materials dynamically, and receiving real-time feedback in an intuitive and efficient manner.

Integrating speech recognition AI presents a transformative opportunity to enhance accessibility, reduce cognitive load, and foster a more engaging learning experience for blind users. Unlike traditional screen readers, which merely vocalize on-screen text, AI-powered speech recognition enables learners to interact with the platform using natural voice commands. This hands-free navigation eliminates the need for extensive keyboard shortcuts or manual exploration, allowing for a more fluid and efficient learning process. Moreover, by incorporating real-time voice interactions, the learning experience shifts from passive listening to active participation, reinforcing comprehension and retention [10]. A fully AI-driven MOOC accessibility model should integrate:

- Advanced Text-to-Speech (TTS): Using natural, expressive voice modulation to enhance clarity and engagement, allowing learners to adjust playback speed, tone, and clarity according to their preferences.
- Voice-Controlled Navigation: Enabling learners to move through lessons seamlessly with voice commands such as "next," "repeat," or "summarize." Additionally, users could request simplified explanations by saying, "Explain this in simpler terms" or "Provide an example."
- Speech-Based Assessments: Incorporating spoken quizzes where learners can answer multiple-choice or open-ended questions through voice input rather than typing, fostering a more interactive evaluation process.
- AI-Powered Conversational Assistants: Providing instant feedback, summarizing lessons, answering content-related questions, and offering personalized study recommendations based on the learner's progress and comprehension.

Furthermore, education is not just about accessing content—it's also about interaction, discussion, and collaboration. Future iterations of the MOOC should include:

- Voice-based discussion forums, where blind learners can participate in audio-based group discussions and interact with instructors and peers.
- Personalized mentorship programs, where audio-based chatbots or real instructors provide one-on-one guidance.
- Peer-to-peer support networks, fostering an inclusive learning community through real-time voice messaging and collaborative learning activities.

There is still room for improvement. Future enhancements should focus on AI-driven text-to-speech improvements, interactive voice navigation, haptic feedback, and community-building features. By integrating these advanced assistive technologies, the platform can provide a more immersive, autonomous, and engaging learning experience for blind users. Through continuous research, innovation, and user feedback, this MOOC can serve as a leading model for inclusive digital education, empowering blind learners with the tools they need to thrive in an online learning environment.

4 Conclusion

This project demonstrates the transformative potential of inclusive digital education for visually impaired learners. By developing an accessible MOOC platform that integrates adaptive technologies and inclusive design principles, it bridges educational gaps and promotes equal opportunities for all learners.

The project's iterative and collaborative approach ensures that it remains responsive to the needs of blind learners and aligned with technological advancements. With a vision of fostering continuous learning and professional development, the inclusive MOOC model serves as a benchmark for accessible digital education, empowering learners to thrive in a digital-first world.

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References

1. Jemni, M., Laabidi, M., Ben Ayed, L.: Accessible E-learning for students with disabilities: from the design to the implementation. In: 10th International Conference on e-Learning, pp. 1–13. Springer, Heidelberg (2014). https://doi.org/10.1007/978-3-642-38291-8_4
2. Królak, A., Chen, W., Sanderson, N.C., Kessel, S.: The accessibility of MOOCs for blind learners. In: 19th International Conference on Human-Computer Interaction, pp. 401–402. ACM, New York (2017). https://doi.org/10.1145/3132525.3134796
3. Park, K., So, H.-J., Cha, H.: Digital equity and accessible MOOCs: accessibility evaluations of Mobile MOOCs for learners with visual impairments. Australas. J. Educ. Technol. **35**, 48–63 (2019). https://doi.org/10.14742/ajet.5521
4. Iniesto, F., Rodrigo, C.: Understanding accessibility in MOOCs: findings and recommendations for future designs. J. Interact. Media Educ. **2024**(1), 20 (2024). https://doi.org/10.5334/jime.897
5. Marques, B.P., Escudeiro, P., Barata, A., Carvalho, P., Sousa, A., Queirós, P.: Assistive technology applied in an inclusive MOOC for the blind. CSEDU. **2**, 100–110 (2019)
6. Escudeiro, P., Galasso, B., Esdras, D., Campos Gouveia, M.: Inclusive MOOC Supported by Assistive Technology. IntechOpen (2023). https://doi.org/10.5772/intechopen.1001121
7. Escudeiro, P., Gouveia, M.C.: Empowering deaf learners: the promise of sign language MOOCs. In: 22nd European Conference on e-Learning, pp. 1–13. Academic Conferences and Publishing Limited (2023)
8. Escudeiro, P., Galasso, B., Teixeira, D., Gouveia, M.C., Escudeiro, N.: Inclusive MOOC—educational content for deaf people, a Portuguese proof of concept. Cypriot J. Educ. Sci. **17**(4), 1285–1303 (2022)
9. Escudeiro, P., Bidarra, J.: Quantitative Evaluation Framework (QEF). Rev. Iberica Sist. Tecnol. Inf., 16–27 (2008)
10. Fu, B., Hadid, A., Damer, N.: Generative AI in the context of assistive technologies: trends, limitations and future directions. Image Vis. Comput. **154**, 105347 (2025). https://doi.org/10.1016/j.imavis.2024.105347